DNA Barcoding Investigation of Fish Authenticity at Whole Foods Market in Boston

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Introduction

Seafood mislabeling, commonly referred to as "fish fraud," is a widespread issue that undermines consumer trust, public health, and sustainable fisheries (Willette et al., 2017). Studies show that mislabeling is especially prevalent among certain high-demand species like salmon, cod, and tuna (Khaksar et al., 2015; Spencer & Bruno, 2019). DNA barcoding has emerged as an effective technique to detect seafood mislabeling by identifying species based on their genetic sequence. As a major retailer that markets itself as a leader in sustainable and ethically sourced foods, Whole Foods claims to offer transparency in product labeling. However, given the national prevalence of fish fraud—even in premium markets (Liou et al., 2020)—it is important to examine whether Whole Foods' fish products are accurately labeled.

Research Question: Are fish products sold at Whole Foods accurately labeled by species, in accordance with their marketed identities?
Hypothesis: The species labels on fish products sold at Whole Foods accurately reflect the true identity of the fish being sold

Methods

1. Collected 3 fish samples (Atlantic salmon, Atlantic cod, tuna) from Whole Foods.
2. Portioned and labeled samples to avoid contamination.
3. Extracted DNA using standard purification techniques.
4. Quantified DNA with a nanodrop spectrophotometer.
5. Performed PCR with samples, positive, and negative controls.
6. Ran gel electrophoresis to check amplification.
7. Sequenced PCR products.
8. Analyzed sequences using BLAST and phylogenetic tools.

Results and Figures

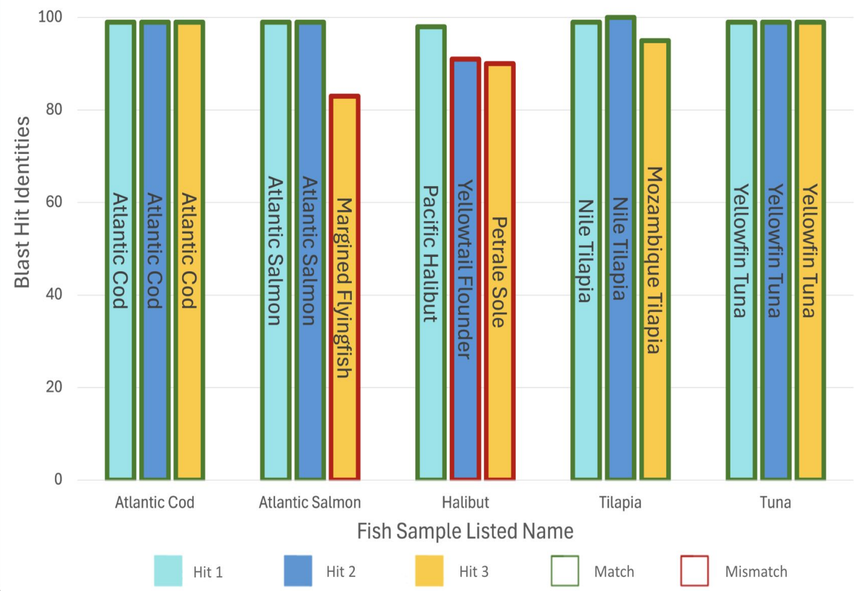

Figure 1. Percent identity of the top three BLAST hits for each fish sample, categorized by predicted species. Bar outlines indicate whether the species identified by BLAST matched the expected species label: green = match, red = mismatch.

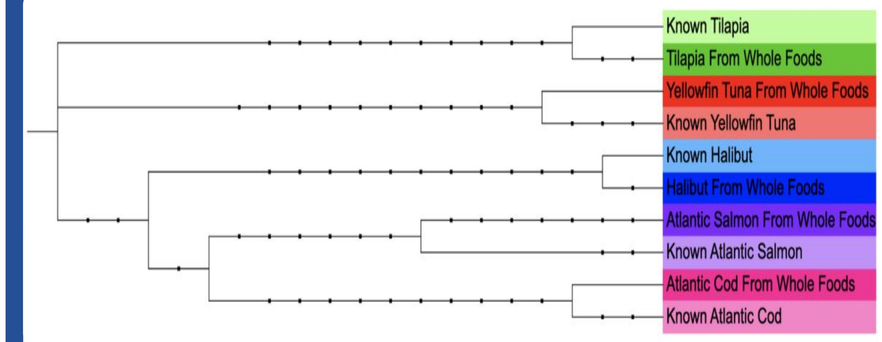

Figure 2. Phylogenetic tree comparing DNA barcoding results of Whole Foods fish sample with known reference species. Samples from my data and peer data clustered tightly with their expected species, supporting correct labeling.

All fish from Whole Foods had top hits matching their labels. Margined Flying Fish, Yellowtail Flounder, Petrale Sole, and Mozambique Tilapia appeared as lower hits but were not top-ranked, indicating no mislabeling.

Discussion

- Hypothesis is supported
  - All three samples had ≥98% identity to their expected species via BLAST.
  - Atlantic salmon, Atlantic cod, and yellowfin tuna each had 99% identity to labeled species.
  - No evidence of mislabeling in any sample.
- Minor off-target BLAST hits observed
  - Lower-ranked matches included species like Margined Flying Fish and Petrale Sole.
  - These were never the top match and likely reflect shared gene regions or noise in database matching.
- Phylogenetic tree analysis supported BLAST results
  - Each sample clustered closely with its expected species.
  - No unexpected placements or outliers in the tree.
- Similar patterns observed in peer samples
  - Other students samples from Whole Foods (e.g., Pacific Halibut, Nile Tilapia) also returned accurate top matches.
  - Indicates consistent labeling quality across multiple Whole Foods samples.

Conclusion

Whole Foods samples were correctly labeled across all tested species
- DNA barcoding confirmed seafood authenticity
- Supports the store's reputation for transparency and traceability

Whole Foods fish can be considered reliably sourced and accurately labeled

References

Khaksar, R., et al. (2015). Unmasking seafood mislabeling in US markets: DNA barcoding as a unique technology for food authentication and quality control. Food Control, 56, 71–76.

Liou, P., et al. (2020). Labeling compliance and species authentication of fish fillets sold at grocery stores in Southern California. Food Control, 112, 107137.

Spencer, E. T., & Bruno, J. F. (2019). Fishy business: red snapper mislabeling along the coastline of the Southeastern United States. Frontiers in Marine Science, 6, 453837.

Willette, D. A., et al. (2017). Using DNA barcoding to track seafood mislabeling in Los Angeles restaurants. Conservation Biology, 31(5), 1076–1085.

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